

Mend-Amar Badral

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I'm a diligent and hard-working student with a passion for applying computational thinking to the most interesting of problems. I aspire to contribute to the advancements in scientific discovery, public understanding of mathematics and open-source technologies. I'm an avid Julia enthusiast and Python programmer.

EDUCATION

Budapest University of Technology and Economics (BME) M.S. in Computer Science Engineering <i>Specialization: Data Science & Artificial Intelligence</i>	Sep, 2025 – June, 2027 Budapest, Hungary
National University of Mongolia (NUM) B.S. in Applied Mathematics Thesis: Evaluating land degradation using image processing methods	Sep, 2021 – June, 2025 Ulaanbaatar, Mongolia

EXPERIENCE

Center of Mathematics Application Research Lab <i>Undergraduate Lab Lead & Research Intern</i>	Oct, 2023 – June, 2025
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- **Lab Leadership:** Selected as Undergraduate Lab Lead in final year; coordinated weekly research progress meetings and facilitated communication between students and faculty.
- **Research assistance:** Participated in a research effort to quantify plant types from drone imagery using image segmentation and statistical methods (thresholding, K-means, Moran's I autocorrelation).
- **Organization:** Managed lab computing resources in a Linux environment and authored technical documentation to standardize researcher onboarding and workflows.
- **Seminar:** Participated in [Study Group Mathematics with Industry 2024](#) in where worked on the problem of liver fat segmentation from microscopy images. In 2025, worked on the problem of [simulating railroad cargo for optimizing costs](#).

PROJECTS

[Deep Learning Foundations and Concepts Book Examples Reproduction](#)

- Reproducing key examples from Bishop's *Deep Learning* textbook to master the mathematical foundations of deep learning from first principles.
- Published reproducible Jupyter and Pluto.jl notebook reports.
- Key topics include: bias of maximum likelihood estimate, curve fitting using polynomials, model complexity and overfitting and regularization

SKILLS

Programming:	Python, Julia, C/C++, R, PyTorch
Technical Communication:	Quarto, L ^A T _E X, Manim
Tools:	Git, Docker, Linux, Bash